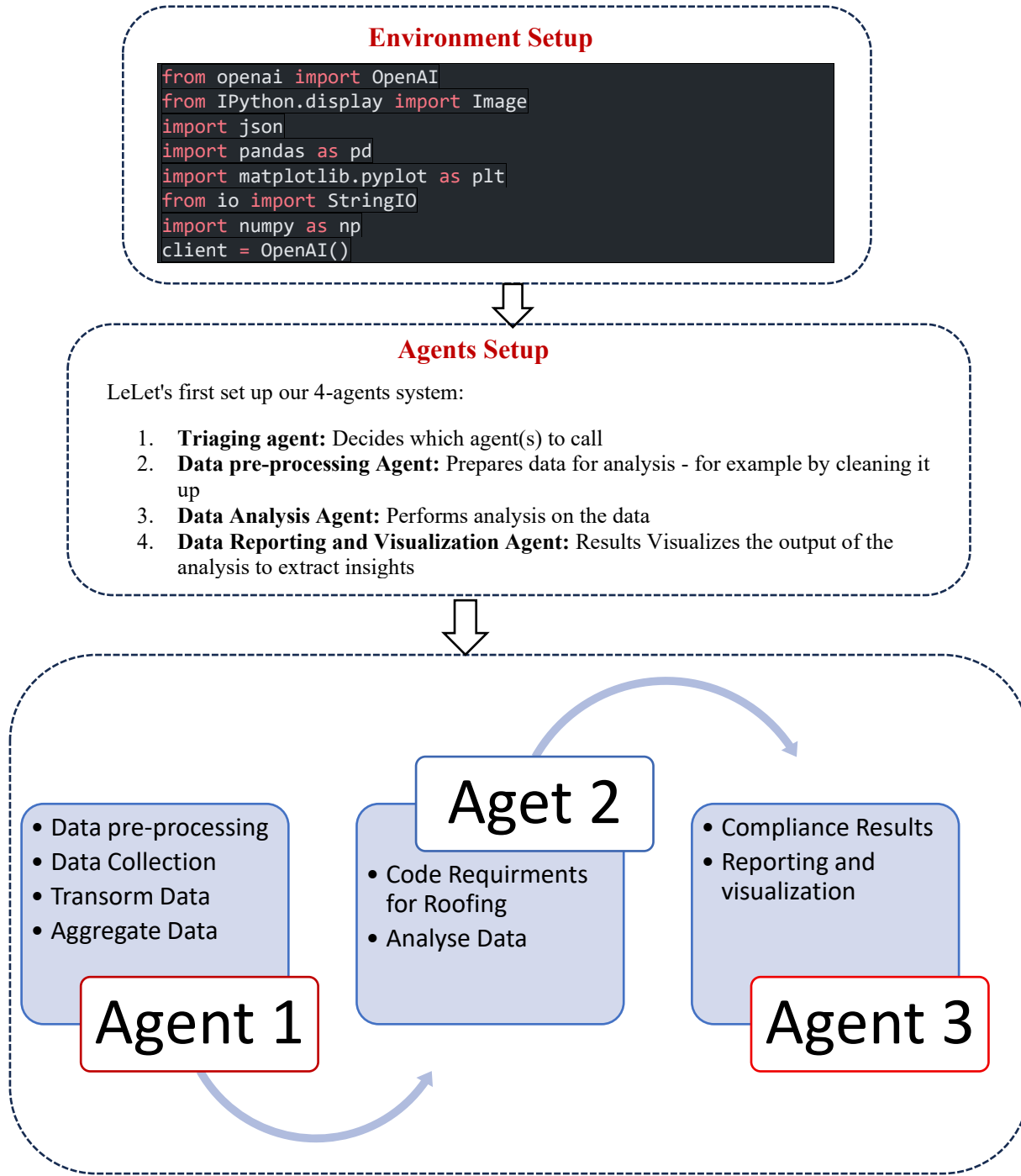


A Multi-Agent System for Roof Design

In an AI application, when using function calling, an increase in the number of functions or tools can lead to reduced performance. To address this, tools can be logically grouped and assigned to specialized "Agents" designed to handle specific tasks or subtasks. This modular approach enhances efficiency and improves overall system performance.

Overview:



We will start by defining the system prompts for each of these agents.

```
triaging_system_prompt = """You are a Triaging Agent. Your role is to assess the user's query and route it to the relevant agents. The agents available are:
```

- Data Processing Agent: Cleans, transforms, and aggregates data.
- Analysis Agent: Performs code compliance analysis for roof design.
- Report Visualization Agent: Creates compliance reports.

```
Use the send_query_to_agents tool to forward the user's query to the relevant agents. Also, use the speak_to_user tool to get more information from the user if needed."""
```

```
processing_system_prompt = """You are a Data Processing Agent. Your role is to collect , transform, and aggregate data using the following tools:
```

- Collect_data
- transform_data
- aggregate_data"""

```
analysis_system_prompt = """You are an Analysis Agent. Your role is to perform roof code compliance analysis using the following tools:
```

- Type of code
- compliance checking analysis
- Further_analysis"""

```
visualization_system_prompt = """You are a Reporting and Visualization Agent. Your role is to create reports for the compliance checking results using:
```

- create_report 1
- create_report 2
- create_report 3"""